Física II UABJ - UFRPE 17/11/2021

Onda sonora num fluido

• Derivada total da pressão $P(x,y,z,t)$

$$dP = a_1 dx + a_2 dy + a_3 dz + a_4 dt$$

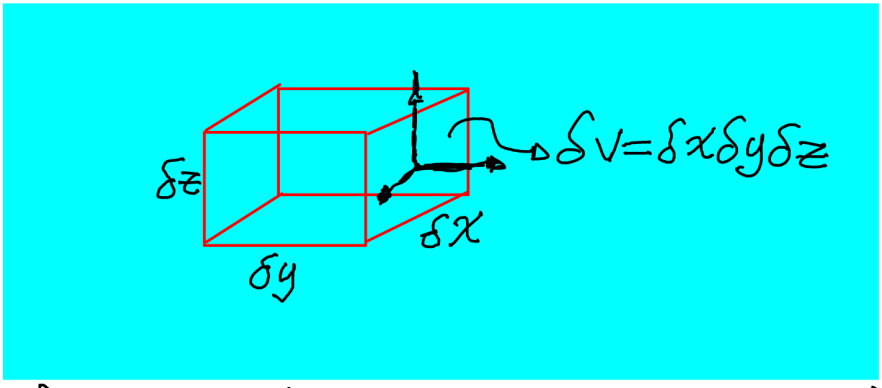
$a_i \quad (i=1,\dots,4)$ $\qquad a_1 = \left.\frac{dP}{dx}\right|_{y=z=t=cte} = \frac{\partial P}{\partial x}$

$$\frac{dP}{dt} = \overbrace{\frac{\partial P}{\partial x}\underbrace{\frac{dx}{dt}}_{v_x} + \frac{\partial P}{\partial y}\underbrace{\frac{dy}{dt}}_{v_y} + \frac{\partial P}{\partial z}\underbrace{\frac{dz}{dt}}_{v_z}} + \frac{\partial P}{\partial t}$$

$$\frac{dP}{dt} = \vec{v}\cdot\vec{\nabla}P + \frac{\partial P}{\partial t} \quad ; \quad \vec{\nabla} = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$$

$$\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$$

$$\boxed{\frac{dP}{dt} = \left(\vec{v}\cdot\vec{\nabla} + \frac{\partial}{\partial t}\right)P}$$



$$d(\delta V) = \underbrace{\delta y\,\delta z}\Big(\underbrace{v_x(x+\delta x,t)\,dt}_{x_f} - \underbrace{v_x(x,t)\,dt}_{x_i}\Big)$$

$$\frac{\partial v_x}{\partial x} = \lim_{\delta x\to 0}\frac{v_x(x+\delta x,t) - v_x(x,t)}{\delta x}$$

$$\underbrace{\delta x\,\frac{\partial v_x}{\partial x}} \simeq v_x(x+\delta x,t) - v_x(x,t)$$

$$d(\delta V) = \underbrace{\delta x\,\delta y\,\delta z}_{\delta V}\left(\frac{\partial v_x}{\partial x} + \delta V\frac{\partial v_y}{\partial y} + \delta V\frac{\partial v_z}{\partial z}\right)dt$$

$$\boxed{d(\delta V) = \delta V\,\vec{\nabla}\cdot\vec{v}} = \delta V\left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)\cdot\left(v_x\hat{i} + v_y\hat{j} + v_z\hat{k}\right)dt$$

$$= \delta V\left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}\right)dt$$

Em resumo

$$\frac{d(\delta V)}{dt} = \delta V\,\underbrace{\vec{\nabla}\cdot\vec{v}}_{\text{Divergencia}}$$

Equação de Continuidade

$$\delta m = \rho\,\delta V = \text{constante}$$

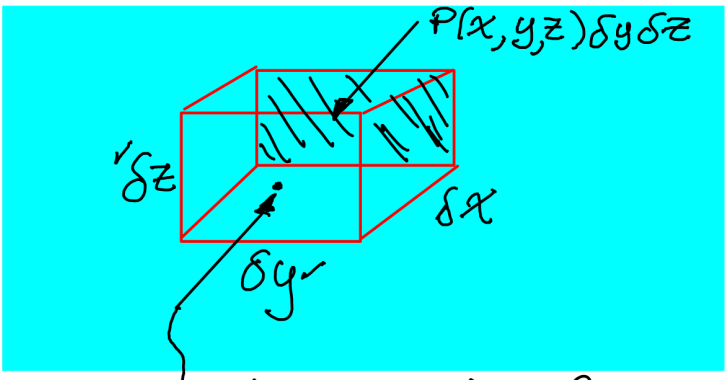
$$d(\delta m) = d(\rho\,\delta V)$$

$$0 = \frac{d\rho}{dt}\delta V + \rho\underbrace{\frac{d}{dt}(\delta V)}_{\vec{\nabla}\cdot\vec{v}\,\delta V}$$

$$\delta V\frac{d\rho}{dt} + \rho\,\vec{\nabla}\cdot\vec{v}\,\delta V = 0$$

$$\left(\frac{\partial\rho}{\partial t} + \vec{v}\cdot\vec{\nabla}\rho + \rho\,\vec{\nabla}\cdot\vec{v}\right)\delta V = 0$$

$$\boxed{\frac{\partial\rho}{\partial t} + \vec{\nabla}\cdot(\rho\,\vec{v}) = 0}$$ Eq. de continuidade



$P(x+\delta x, y, z)\,\delta y\,\delta z$

$$\Sigma\,\delta F_x = \left(P(x,y,z) - P(x+\delta x, y, z)\right)\delta y\,\delta z$$

$$\frac{\partial P}{\partial x} = \lim_{\delta x \to 0} \frac{P(x+\delta x, y, z) - P(x, y, z)}{\delta x}$$

$$-\delta x \frac{\partial P}{\partial x} = P(x, y, z) - P(x+\delta x, y, z)$$

$$\delta F_x = -\frac{\partial P}{\partial x} \delta x \delta y \delta z = -\frac{\partial P}{\partial x} \delta V$$

$$\delta \vec{F} = \delta F_x \hat{i} + \delta F_y \hat{j} + \delta F_z \hat{k}$$

$$\delta \vec{F} = -\delta V \left(\frac{\partial P}{\partial x} \hat{i} + \frac{\partial P}{\partial y} \hat{j} + \frac{\partial P}{\partial z} \hat{k} \right)$$

$$\boxed{\delta \vec{F} = -\vec{\nabla} P \delta V}$$

$\vec{f}$: força externa por unidade de volume

$$\delta \vec{F}_{Total} = -\vec{\nabla} P \delta V + \vec{f} \delta V$$

Usando a segunda lei de Newton

$$\Sigma \vec{F} = m\vec{a} \qquad m = \delta m = \rho \delta V \; ; \; \vec{a} = \frac{d\vec{v}}{dt}$$

$$\left(-\vec{\nabla} P + \vec{f}\right) = \rho \underbrace{\frac{d\vec{v}}{dt}}$$

$$\left(-\vec{\nabla} P + \vec{f}\right) = \rho \left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \vec{\nabla} \vec{v} \right)$$

$$\boxed{\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \vec{\nabla} \vec{v} = \frac{\vec{f}}{\rho} - \frac{\vec{\nabla} P}{\rho}}$$

Eq. de Euler para um fluido

$$P = P_0 + P' \qquad P' \ll P$$

$$\rho = \rho_0 + \rho' \qquad \rho' \ll \rho$$

$$\vec{f}_0 = \vec{\nabla} P_0$$

$$\frac{\partial \vec{v}}{\partial t} + \vec{v}\cdot\vec{\nabla}\vec{v} = \frac{\vec{f}_0}{\rho_0+\rho'} - \frac{\vec{\nabla}(P_0+P')}{\rho_0+\rho'}$$

$$\frac{1}{\rho_0+\rho'} = \frac{1}{\rho_0\left(1+\frac{\rho'}{\rho_0}\right)} = \frac{1}{\rho_0}\left(1+\frac{\rho'}{\rho_0}\right)^{-1}$$; Lembrar $(1+x)^n \simeq 1+nx+n(n-1)x^2+\ldots$

$$\simeq \frac{1}{\rho_0}\left(1-\frac{\rho'}{\rho_0}\right)$$

$$\frac{\partial \vec{v}}{\partial t} + \underbrace{\vec{v}\cdot\vec{\nabla}\vec{v}}_{\to 0} \simeq \frac{\vec{f}_0}{\rho_0}\left(1-\frac{\rho'}{\rho_0}\right) - \frac{\vec{\nabla}P_0+\vec{\nabla}P'}{\rho_0}\left(1-\frac{\rho'}{\rho_0}\right)$$

$$\vec{f}_0 = \vec{\nabla}P_0$$

$$\boxed{\frac{\partial \vec{v}}{\partial t} \simeq -\frac{\vec{\nabla}P'}{\rho_0}}$$

Usando nossa approximação na equação de continuidade

$$0 = \frac{\partial \rho'}{\partial t} + \vec{\nabla}\cdot\left((\rho_0+\rho')\vec{v}\right)$$

$$0 \simeq \frac{\partial \rho'}{\partial t} + \vec{\nabla}\cdot\underbrace{(\rho_0\vec{v})}$$

$$0 = \frac{\partial \rho'}{\partial t} + \rho_0\,\vec{\nabla}\cdot\vec{v}$$

Compressibilidade adiabática K e seu inverso $B = \frac{1}{K}$

$$K_s = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)\Big|_{s=cte} = \frac{1}{B_s}$$

$$\rho = m/V \;;\; V = m/\rho$$

$$\frac{1}{B_s} = K_s = -\frac{1}{m/\rho}\left(\frac{\partial m/\rho}{\partial P}\right)_s = -\rho\frac{\partial}{\partial P}\left(\frac{1}{\rho}\right)_s = -\rho\left(\frac{-1}{\rho^2}\frac{\partial \rho}{\partial P}\right)_s$$

$$\frac{1}{B_s} = K_s = \frac{1}{\rho}\left(\frac{\partial \rho}{\partial P}\right)_s$$

$$\frac{1}{B_s} = K_s = \frac{1}{\rho_0}\frac{\rho'}{P'} \Rightarrow \boxed{\rho' = \frac{\rho_0 P'}{B_s}}$$

Retornando para a equação de Continuidade

$$\rightarrow \frac{\partial \rho'}{\partial t} + \rho_0 \vec{\nabla}\cdot\vec{v} = 0$$

$$\frac{\partial}{\partial t}\left(\frac{\rho_0 P'}{B_s}\right) + \rho_0 \vec{\nabla}\cdot\vec{v} = 0$$

$$\rightarrow \frac{1}{B_s}\frac{\partial P'}{\partial t} + \vec{\nabla}\cdot\vec{v} = 0$$

Derivando em função do tempo

$$\frac{1}{B_s}\frac{\partial^2 P'}{\partial t^2} + \vec{\nabla}\cdot\frac{\partial \vec{v}}{\partial t} = 0$$

usando a eq. de Euler $\frac{\partial \vec{v}}{\partial t} = -\frac{\vec{\nabla} P'}{\rho_0}$

$$\frac{\partial^2 P'}{\partial t^2} - \frac{B_s}{\rho_0}\vec{\nabla}\cdot\vec{\nabla}P' = 0$$

$$\boxed{\frac{\partial^2 P'}{\partial t^2} - \frac{B_s}{\rho_0}\nabla^2 P' = 0} \qquad \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$\rightarrow$ Equação tridimensional de onda

Velocidade de propagação $v^2 = \frac{B_s}{\rho_0}$

$$\boxed{\frac{\partial^2 P'}{\partial t^2} - v^2\nabla^2 P' = 0}$$

A velocidade de propagação num gas ideal

- processo adiabático $PV^\gamma = cte$ $\quad \gamma = c_p/c_v$

$$\log(ab) = \log(a) + \log(b)$$

$$\ln(PV^\gamma) = \ln P + \gamma \ln V = c_2 \quad \text{ao derivar}$$

$$\frac{dP}{P} + \gamma\frac{dV}{V} = 0 \Rightarrow \frac{dP}{P} = -\gamma\left(\frac{dV}{V}\right) \Rightarrow \frac{-dP}{dV/V} = \gamma P$$

$$B_s = -\frac{dp}{dV/V} = \gamma P$$

Velocidade de propagação $v = \sqrt{\frac{B_s}{\rho_0}} = \sqrt{\frac{\gamma P}{\rho_0}}$

$PV = nRT$; $P = \rho \frac{RT}{M}$; $\frac{P}{\rho} = \frac{RT}{M}$

$v = \sqrt{\frac{\gamma RT}{M}}$ => A velocidade de propagação de uma onda sonora num gas ideal.